Cerebellum

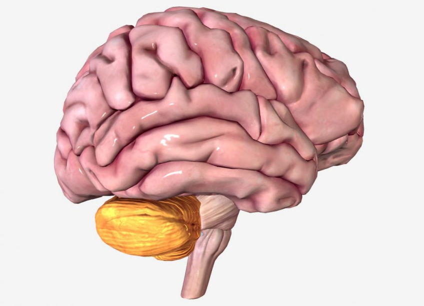
Anatomy, function and microstructure

Computational Neuroscience
University of Bristol
Slides adapted from L Aitchison
M Rule

Learning outcomes:

- Cerebellar gross anatomy, microstructure, function, and the link to supervised learning

Cerebellum



"little brain"

80% of the brain's neurons (Azevedo et al. 2009)

- of which most are cerebellar *granule cells*
- ~70 billion cerebellar granule cells (Lange 1975)

Damage → ataxia & cognitive deficits

Function:

- Forward modelling
  - Reshape motor commands from cortex to remove predictable errors
- Doya (1999,2000)
  - The Cerebellum is the brain's supervised learning module
- Cognitive roles
  - Ito (2008) *Control of mental activities by internal models in the cerebellum*

Cerebellar motor deficits

A standard test of cerebellar function is to reach with the tip of the finger for a target at arm's length: A healthy person will move the fingertip in a rapid straight trajectory, whereas a person with cerebellar damage will reach slowly and erratically, with many mid-course corrections.

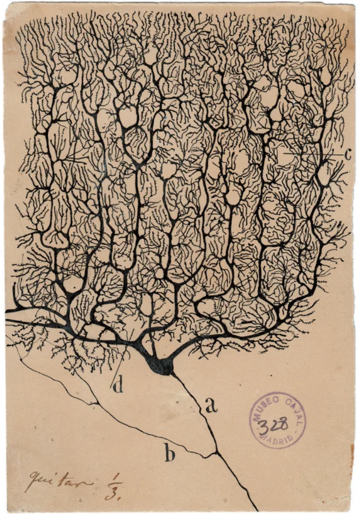
Motor deficits with cerebellar damage:

- speech deficits (slow speech, slurring)
- oculomotor deficits (e.g. instability of gaze)
- tremor
- difficulties in balance causing swaying + broad based stance

Cerebellar Ataxia, André Thomas (1012) Nervous and mental disease monograph series, issues 12, via Wikimedia

Cerebellar Purkinje cells (PCs)

- Dendrites are planar!
- 100,000 input synapses (far more than most other cell types)
- Typical (simple-spike) firing rate = 50 Hz!



Cajal

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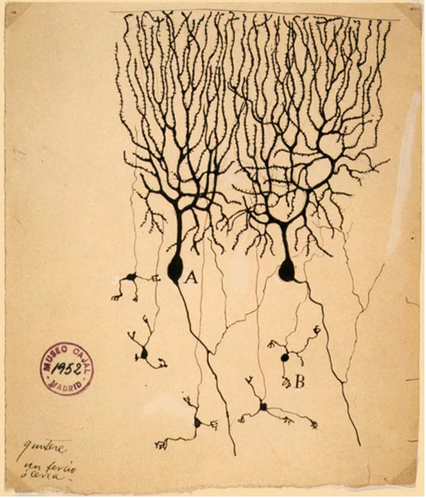

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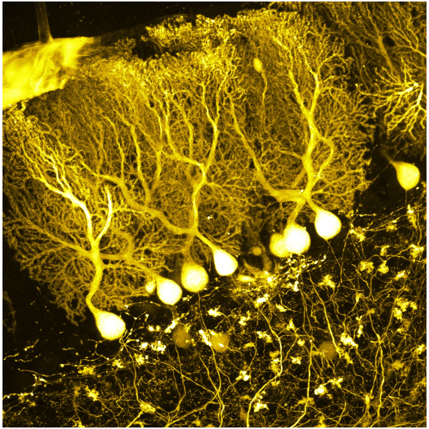

BrainsRusDC

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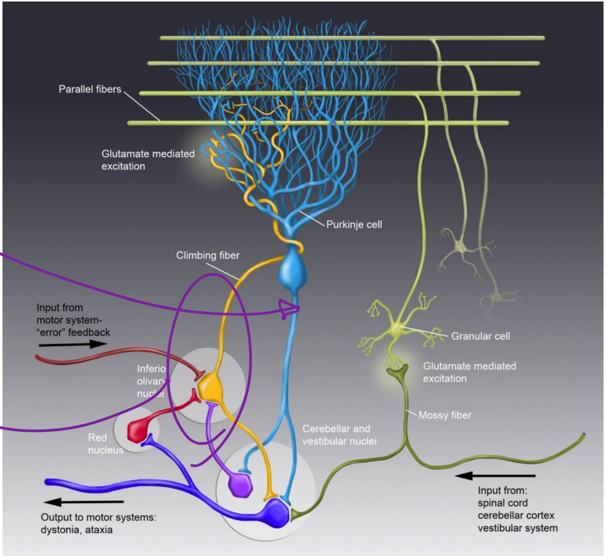
Maryann Martone, CCDB/NCMIR/UCSD

Cerebellar Microcircuit

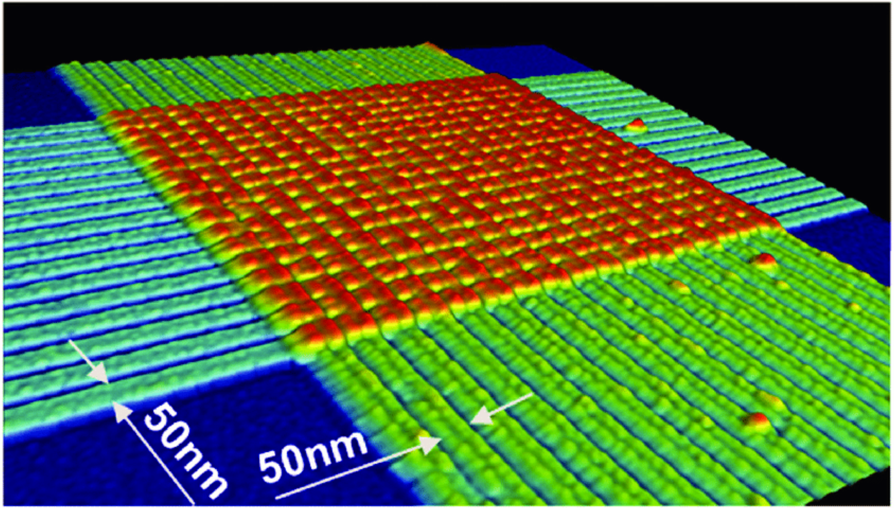
cable that carries ε (or something like it) computing a fwd model to correct errors



generating features

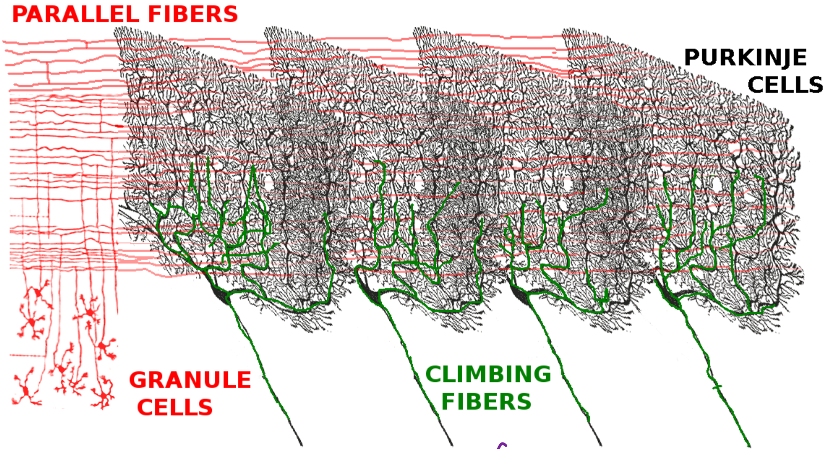
Apps & Garwicz (2005)

Cerebellar Microcircuit



memristor crossbar array

Cerebellar Microcircuit



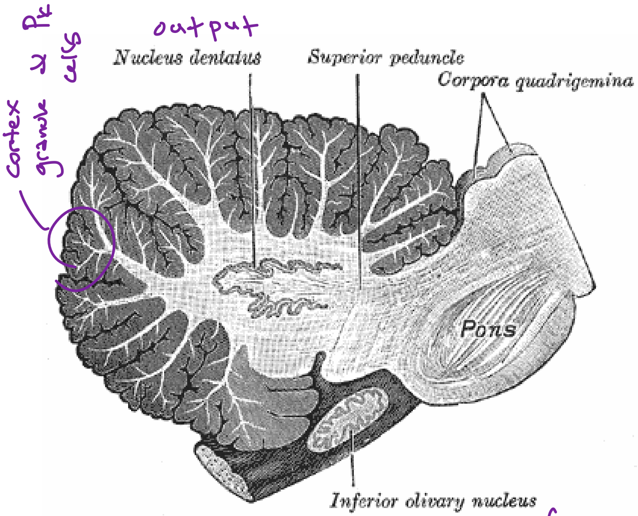
C Houghton

(1 fiber per Pk cell

Inferior olive → climbing fibres

Pons (others) → mossy fibres

Purkinje cell output → dentate nucleus (others)



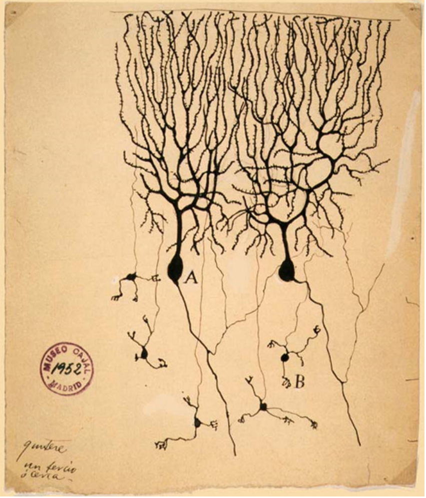
Gray's anatomy

Climbing fiber

ε

Granule cell inputs

- "B" in the image
- Most numerous cell type in the brain (70 billion cells)
- 1-7 synapses from mossy fibres → sparse, decorrelated representation
- Outputs → "parallel fibres"
- Granule cells themselves take input through the mossy fibres from all over the brain



Cajal

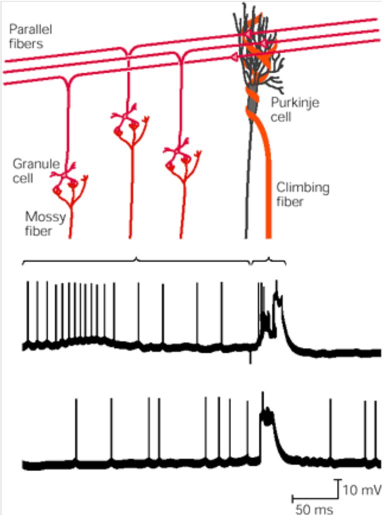
Climbing fibre error signals

- There is one climbing fibre for each Purkinje cell.
- Climbing fibres wrap around the PC and synapse very, very strongly.
- Climbing fibres fire at a low rate ~1 Hz
- Climbing fibres originate in the inferior olive.

Wilson et al(2019)

Simple and complex spikes

- Standard PC inputs cause high-frequency (∼50 Hz) simple-spikes.
- Climbing fibre inputs causes low-frequency complex spike (∼1 Hz)
- Complex spikes signal errors, and cause LTD (i.e. indicate simple-spike output was too high)



Martinez et al. (1971)

Cerebellum is assumed to do supervised learning

Climbing fibre spikes

- Indicate a negative error
- Cause long-term depression of recently active granule→Purkinje synapses

$$\hat{y} = \sum_i w_i x_i$$

$\hat{y}$ Purkinje cell simple spikes

x_i Parallel fibre / granule-cell inputs

w_i granule-cell $\rightarrow$ Purkinje cell synapses

weight update:

$$\Delta w_i \propto \varepsilon \cdot x_i$$

it's OK!

~~flipped~~ ?

$$\varepsilon = \cancel{-} \underbrace{\min\{0, y^* - \hat{y}\}}$$

− climbing fibre errors

Do mrule's slides have typo?

Let's check, first principles:

Delta rule is $\Delta w \propto (y^* - y)x$

Climbing fiber can only decrease w & granule cell (parallel fibre) "x" is always positive. So, could implement δ rule when $\Delta w \propto (y^* - y)x < 0$

i.e. $y^* - y = \varepsilon < 0$.

we typically call all spiking input (+) so climbing fiber can't be $\varepsilon < 0$, rather is

ReLu

$$\boxed{-\varepsilon = \max[0, \hat{y} - y^*]}$$

or

$$\boxed{\varepsilon = \min[0, y^* - \hat{y}]}$$

it's OK!

If climbing fibres only cause LTD, what causes LTP?

- Calcium spikes in the dendrites (with sufficient input)
- Recall the *dendritic integration* slides, and view of neuron as $n \geq 2$-layer net

Adapted from L Aitchison

(i.e. climbing-fibre / complex spike indicate output was too large)

Cerebellum self-test

80% (but exact # not on exam)

The Cerebellum

- Contains ~~70%~~ of the brain's neurons,
- Computes forward models (motor and cognitive)
- Damage causes Ataxia + cognitive deficits

The microcircuit

- mossy fibers bring inputs from the brain to cerebellar granule cells
- 70 billion granule cells give rise to parallel fibers
- Purkinje cells have up to 100K parallel fibre inputs
- Each Purkinje cell received error feedback from a single climbing fiber from the inferior olive

The cerebellum performs Supervised learning

- Climbing fibre spikes signal a error
- and cause long-term depression of granule→Purkinje cell synapses

Cerebellum self-test

The Cerebellum

- Contains 80% of the brain's neurons,
- Computes forward models (motor and cognitive)
- Damage causes ataxia, various motor symptoms

The microcircuit

- Mossy fibres bring inputs from the brain to cerebellar granule cells
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The cerebellum performs supervised learning

- Climbing fibre spikes signal a negative prediction error
- and cause long-term depression of granule$\rightarrow$Purkinje cell synapses

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